

PROMPT: 68-year-old male with progressive **dyspnea**, **orthopnea**, and leg edema. Echo shows EF **35%**, consistent with **heart failure**. ... mild pulmonary congestion; elevated creatinine suggestive of cardiorenal syndrome.

Task: Predict ICD-10 codes from a large set (10000+ labels).

I50.22 - Chronic systolic HF J18.9 - Pneumonia I42.9 - Cardiomyopathy
N18.3 - CKD (stage 3) I25.10 - Coronary artery disease N17.9 - Acute kidney failure
J44.9 - COPD ... +9000 unrelated codes

Later Fine-Grained Chain-of-Thought Reasoning

EF 35% confirms **systolic dysfunction (HFrEF)** - rules out **HFpEF** (EF typically $\geq 50\%$). Chronic orthopnea + peripheral edema pattern favors **chronic systolic HF** over acute myocarditis or flash pulmonary edema from other causes. Elevated creatinine in low-output state supports cardiorenal syndrome (secondary to HF) rather than primary renal disease as driver of fluid overload.

Contrastive differentiation: true vs near-miss candidates

HFrEF (EF 35%)

vs **HFpEF (EF $\geq 50\%$)**

Chronic systolic HF

vs **Acute myocarditis / flash edema**

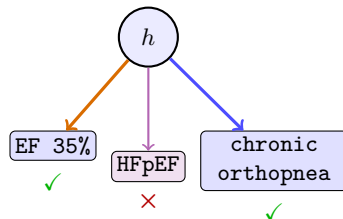
Cardiorenal syndrome (secondary)

vs **Primary renal failure**

Mechanistic Question—Phase 2

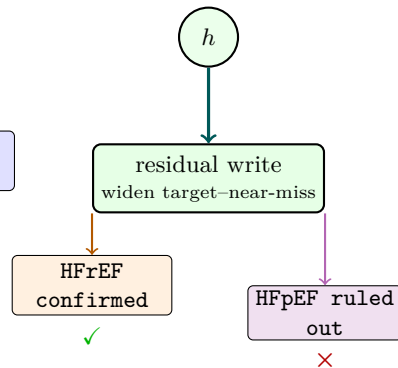
When a later CoT token performs fine-grained contrastive reasoning, what is an attention head doing internally?

later CoT token: "HFrEF (EF 35%)"



What did it attend to?

iterative contrastive attention:
boost own prior true shortlist (✓)
suppress near-miss candidates (✗)



What did it write?

margin shaping:
boost target (✓)
separate near-miss (✗)